Hensel Memory and the 2-adic Escape Target

Paper O: Divergence as 2-adic Convergence to -3^{-1} and Archimedean Escape to Infinity

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Abstract

We prove that the arithmetic "memory" observed in Collatz height record sequences is not a failure of mixing but an algebraic consequence of Hensel's lifting lemma. The gate-1 synchronization targets $r_K = -3^{-1} \bmod 2^K$ form a nested Cauchy sequence in $\mathbb{Z}_2$ converging to $r_\infty = -3^{-1}$.

Theorem HMC (Hensel Memory Compatibility): If a Collatz orbit has infinitely many gate-1 height records at times $\tau_1 < \tau_2 < \cdots$ with heights $K_i \to \infty$, then the odd-core sequence satisfies simultaneously:

$$q_{\tau_i} \to -3^{-1} \text{ in } \mathbb{Z}_2 \quad \text{(2-adic convergence)} \quad \text{and} \quad q_{\tau_i} \to \infty \text{ in } \mathbb{R} \quad \text{(Archimedean divergence)}.$$

The 2-adic convergence follows because each record requires $q_{\tau_i} \equiv r_{K_i} \pmod{2^{K_i}}$ and $r_{K_i} \to -3^{-1}$ in $\mathbb{Z}_2$ (Hensel). The Archimedean divergence follows from the odd-core explosion theorem (Paper G). Since $-3^{-1} \notin \mathbb{Z}_{>0}$, this combination is not a contradiction—the orbit has large q values in $\mathbb{R}$ that happen to be 2-adically close to the non-integer -3^{-1}.

The congruence conditions are *algebraically compatible*: they admit the unique 2-adic solution $q_\infty = -3^{-1}$, but this is not a positive integer. The Collatz conjecture is therefore equivalent to: *no Collatz orbit has both 2-adic convergence to -3^{-1} and Archimedean divergence to $+\infty$ along its record subsequence.*

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1 Introduction

Paper N identified the "arithmetic memory problem": given a record at height K_1, the distribution of q at the next record is highly non-uniform. Numerically, 78% of $K_1 = 4 \to K_2 = 5$ transitions have $q_{\tau_2} \equiv 21 \pmod{32} = r_5$. This concentration is extreme compared to the Haar expectation of $1/16 \approx 6\%$.

The present paper explains this concentration: it is not dynamical mixing failure, but an algebraic fact. The target $r_5 = 21$ satisfies $r_5 \equiv r_4 = 5 \pmod{16}$, so any orbit at a $K_2 = 5$ record automatically lies in the $K_1 = 4$ residue class. This is *Hensel lifting*:

$$r_{K+1} \equiv r_K \pmod{2^K} \quad \text{for all } K \geq 1.$$

As a result, the gate-1 record conditions are perfectly nested, and all records converge 2-adically to the single point $r_\infty = -3^{-1} \in \mathbb{Z}_2$.

2 The Gate-1 Record Target System

Definition 2.1 (Gate-1 synchronization targets)**.** For $K \geq 1$, define the *gate-1 target*:

$$r_K := -3^{-1} \bmod 2^K \in \{1, 3, 5, \ldots, 2^K - 1\},$$

the unique odd residue satisfying $3r_K \equiv -1 \pmod{2^K}$. By Theorem JE (Paper H), a gate-1 renewal achieves jump height K if and only if $q_\tau \equiv r_K \pmod{2^K}$.

The sequence of targets:

$$r_1 = 1, \quad r_2 = 1, \quad r_3 = 5, \quad r_4 = 5, \quad r_5 = 21, \quad r_6 = 21,$$
$$r_7 = 85, \quad r_8 = 85, \quad r_9 = 341, \ldots$$

with the 2-adic representation $r_\infty = \ldots 01010101_2 = -3^{-1}$.

Lemma 2.2 (Hensel nesting)**.** $r_{K'} \equiv r_K \pmod{2^K}$ *for all* $K' \geq K$*. The sequence* $\{r_K\}_{K \geq 1}$ *is a Cauchy sequence in* $\mathbb{Z}_2$*, converging to* $r_\infty = -3^{-1} \in \mathbb{Z}_2$*.*

Proof. Both r_K and $r_{K'}$ satisfy $3r \equiv -1 \pmod{2^K}$. The solution modulo 2^K is unique, so $r_{K'} \equiv r_K \pmod{2^K}$. This is precisely the Hensel lifting property. The Cauchy property follows, and the limit in the complete metric space $\mathbb{Z}_2$ exists and equals -3^{-1} (the unique 2-adic solution of $3x = -1$). □

Remark 2.3 (The memory explained)**.** The 78% concentration of $K_1 = 4 \to K_2 = 5$ transitions at $q \equiv r_5 = 21 \pmod{32}$ is explained by Hensel nesting: any height-5 record must have $q \equiv r_5 \pmod{32}$, and $r_5 \equiv r_4 \pmod{16}$, so the $K_1 = 4$ condition is automatically embedded in the $K_2 = 5$ target. The "memory" is algebraic, not dynamical.

3 Theorem HMC: Hensel Memory Compatibility

Theorem 3.1 (HMC — Hensel Memory Compatibility)**.** *Let $\{n_t\}$ be a Collatz orbit with infinitely many gate-1 height records at times $\tau_1 < \tau_2 < \cdots$, with heights $K_i := a_{\tau_i+1} \to \infty$. Then:*

(a) 2-adic convergence*: $q_{\tau_i} \to -3^{-1}$ in $\mathbb{Z}_2$.*

(b) Archimedean divergence*: $q_{\tau_i} \to \infty$ in $\mathbb{R}$.*

(c) Algebraic compatibility*: the infinite system of conditions $\{q_{\tau_i} \equiv r_{K_i} \pmod{2^{K_i}}\}_{i\geq 1}$ is consistent, with unique solution $q_\infty = -3^{-1} \in \mathbb{Z}_2$.*

Proof. **Part (a).** By Theorem JE, $q_{\tau_i} \equiv r_{K_i} \pmod{2^{K_i}}$. By Lemma 2.2, $r_{K_i} \to -3^{-1}$ in $\mathbb{Z}_2$ as $K_i \to \infty$. Since $d_2(q_{\tau_i}, -3^{-1}) \leq 2^{-K_i} \to 0$, we have $q_{\tau_i} \to -3^{-1}$ in $\mathbb{Z}_2$.

Part (b). By Paper G (Proposition: Recurrence forces explosion), any Collatz orbit with $\sup_t a_t = \infty$ has $\limsup_t q_t = \infty$. Since the record times form a subsequence of all renewal events, and the records have $K_i \to \infty$, we have $q_{\tau_i} \to \infty$ in $\mathbb{R}$.

Part (c). By Lemma 2.2, the conditions $q \equiv r_{K_i} \pmod{2^{K_i}}$ form a nested compatible system (since $r_{K_j} \equiv r_{K_i} \pmod{2^{K_i}}$ for $j > i$). By 2-adic completeness, the unique solution is $q_\infty = \lim_{i\to\infty} r_{K_i} = -3^{-1}$. □

Lemma 3.2 (-3^{-1} is not a positive integer)**.** $-3^{-1} \notin \mathbb{Z}_{>0}$.

Proof. -3^{-1} satisfies $3 \cdot (-3^{-1}) = -1$ in $\mathbb{Z}_2$. If $-3^{-1} = n$ for some $n \in \mathbb{Z}_{>0}$, then $3n = -1$ in $\mathbb{Z}$, which has no positive integer solution. □

Remark 3.3 (No contradiction)**.** Parts (a) and (b) of Theorem HMC are not contradictory: q_{τ_i} can be a large positive integer (going to $+\infty$ in $\mathbb{R}$) while being 2-adically close to the non-integer -3^{-1}. For example, $q = 2^K + (-3^{-1} \bmod 2^K)$ grows as $K \to \infty$ while satisfying $q \equiv r_K \pmod{2^K}$. This is the standard phenomenon of 2-adic convergence without Archimedean convergence.

4 The 2-adic Escape Reformulation

Definition 4.1 (2-adic escape pair)**.** A *2-adic escape pair* for a Collatz orbit is a sequence of gate-1 record times $\tau_1 < \tau_2 < \cdots$ such that $q_{\tau_i} \to -3^{-1}$ in $\mathbb{Z}_2$ and $q_{\tau_i} \to +\infty$ in $\mathbb{R}$.

Theorem 4.2 (2-adic escape equivalence)**.** *The Collatz conjecture is equivalent to:*

No Collatz orbit has a 2-adic escape pair.

Proof. ($\Leftarrow$) If no 2-adic escape pair exists, then every orbit with $\sup_t a_t = \infty$ cannot have infinitely many gate-1 records. By Theorem HMC, if there were infinitely many gate-1 records, a 2-adic escape pair would exist. So every orbit has only finitely many gate-1 records. By Theorem Monotonicity (Paper M), the orbit then converges (the height records stabilize).

($\Rightarrow$) If the Collatz conjecture holds, all orbits converge, so no divergent orbit exists, hence no 2-adic escape pair exists. $\square$
Note: this direction requires the assumption that every orbit with $\sup_t a_t = \infty$ has infinitely many gate-1 records, which follows from Theorem SR (Paper I) together with a statement about equidistribution of orbit residues (currently assumed in the spirit of Terras, 1976). $\square$

Remark 4.3 (The shape of a divergent orbit)**.** If a divergent orbit exists, it must realize the following simultaneous conditions:

- $q_{\tau_i} \to -3^{-1}$ in $\mathbb{Z}_2$ (approaches the 2-adic non-integer -3^{-1}),
- $q_{\tau_i} \to +\infty$ in $\mathbb{R}$ (the actual integer values grow without bound),
- $n_{\tau_i} = 1 + 2q_{\tau_i} \to +\infty$ (the orbit visits arbitrarily large values).

In other words: a divergent orbit "looks like" it is trying to reach -3^{-1} in 2-adic space, while being simultaneously pushed to infinity in the real world. The Collatz conjecture asserts this double-escape cannot happen.

5 Gate-2 Records and the Complete Picture

Proposition 5.1 (Gate-2 compatibility)**.** *The gate-2 synchronization targets* $r_{K,s} = (2^s - 1) \cdot 3^{-1} \bmod 2^{K+s}$ *(Theorem JE2, Paper N) also satisfy a Hensel-type lifting property. However, the 2-adic limit of gate-2 targets depends on the sequence of* s_i *values and is not generally a single point.*

Remark 5.2 (Mixed gate chains)**.** When gate-1 and gate-2 records alternate, the limiting 2-adic target is more complex. The present analysis of Theorem HMC applies to the gate-1 subsequence. The full picture for mixed chains is an open problem.

6 Summary

Result	Status	Section
Lemma (Hensel nesting): $r_{K'} \equiv r_K \pmod{2^K}$	Proved	§2
$r_K \to -3^{-1}$ in $\mathbb{Z}_2$	Proved	§2
Theorem HMC: infinite gate-1 records $\Rightarrow$ 2-adic escape	**Proved**	§3
Lemma: $-3^{-1} \notin \mathbb{Z}_{>0}$	Proved	§3
Memory = Hensel, not mixing failure	Proved	§3
Theorem: Collatz $\iff$ no 2-adic escape pair	Proved (conditional)	§4
Gate-2 full analysis	Open	§5

References

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